Hero inverted-U — three model layers (v1)

Date: 2026-07-20
Audience: Charles + Alex read-through
Canonical hero folder: `sports/datasets/mbb/exports_inverted_u_v0/alex_side_by_side_v0/`
Related: `05_Model_Nesting_Note_v1.md`, `30_Alex_Gates_inform_status_outline.md`

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1. Locked hero estimand (fixed target for all tests)

All Layer A/C comparisons use **this** empirical spec unless explicitly noted as robustness.

Field	Value
Outcome	`Y_draft` (ever drafted), mean rate per bin
X-axis	`poolq_loo` (LOO mean teammate `perf`)
perf	ESPN box **ppm**, **z-scored within collegiate season**
Bins	**16 quantile** (`poolq_binning='quantile'`)
Winsor	`poolq_loo` at **(0.01, 0.99)** before binning
min_minutes	**20** (applied before LOO when rebuilding from box)
Panel source	`use_prebuilt_panel_csv=False` for canonical rebuild path
Draftee filter	**Off** (`restrict_teams_by_draftees=False`)
Seasons	**2011–2021**
Sample (Jul 2026 lock)	**62,180** player-seasons; **1,134** drafted

File naming slug: `empirical_ppm_poolq_loo_16quantile_winsor0199_min20_2011`

Shape (qualitative): rise through middle ventiles → plateau bins 12–15 (~2.3–2.6%) → **dip at bin 16 only** (~1.2%).

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2. Three layers — do not merge into one "model"

Layer	Object	Repo anchor	What a "pass" proves	What it does **not** prove
A. Phenomenological	`Y ~ β₀ + β₁·poolq_loo + β₂·poolq_loo²`	`530` / `panel_build.draft_poolq_quadratic_coeffs`	Inverted curvature in the **real panel** on the hero estimand	Mechanism; that NBA uses this score
B. Structural (theory)	`L_net = B(·) − D(·)`; Alex `S_i = A_i − λ·L_C` in selection	Nesting note §2–3	Why help and hurt can coexist; **D enters selection**	Separate estimation of B(Q) and D(Q) in v1
C. Generative (sim)	Soft assign → score → top-K	`538D` CELL 10–12, `tier1_generative_eda.py`	**Congestion in the score** changes selection curves; **talent-only fails**	Bin-for-bin reproduction of empirical `poolq_loo` ventiles

Alex side-by-side (v1): Layer **A hero PNG** + Layer **C sim PNG** on **same Y** (advancement rate), **honest X** (empirical LOO quality vs sim LOO bins), plus limitation sentence below.

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3. How each layer is tested

Layer A — regression on real rows

- Filter panel to hero estimand → OLS `Y_draft ~ 1 + poolq_loo + poolq_sq`.
- **Pass:** `β₂ < 0` (concave); optional overlay tracks ventile bin means.
- **Not required:** R^2, causal interpretation, or match to sim.

Layer B — prose + ingredient knockouts

- No single estimated formula in v1.
- **Pass:** Narrative consistency + sim shows λ=0 (ability-only) **fails** the generative congestion story.

Layer C — simulation

- Draw synthetic league → assign rosters → rank by score → select top **K** → bin by `poolq_loo` (16 quantile).
- **Knockout A (talent-only):** `score_mode='ability'` → **monotone** selection rate on `poolq_loo` (bin 1 ~0.7% → bin 16 ~28%); no elite dip.
- **Knockout B (congestion):** `score_mode='loo_gap_plus_ability'`, `loo_pool_l_mode='crowding_smooth'`, `w=0.5` → elite-tail **compression** (bin 16 ~16% vs ~28% talent-only); not pointwise hero match.

- **Script:** `sports/scripts/hero_model_reset_bundle.py` writes CSV + side-by-side PNG.

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4. Simplification strategies (both are valid)

Strategy	Use for	Procedure
Bottom-up	Layer A	Start quadratic → decompose Q only at Rung 2.5 (`poolq_loo` vs `crowding_smooth` in CELL 5d)
Top-down	Layer C	Full generative stack → zero λ / ability-only / drop assignment knobs → record what breaks

Use **bottom-up first** for the hero; **top-down** for 538D mechanism strip-down.

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5. Simulation plumbing vs core claim

Knob	Role	v1 gate?
λ / w on L_C in score	Core mechanism	**Yes**
Top-K selection	Scarcity	**Yes (conceptual)**
Soft assignment (τ, T_j)	Realistic pools	Helpful, not theorem
Assortative sort-and-chop overlay	Diagnostic (Plot A)	No
Faithful-538 empirical team sweeps	Realism stretch	**No — see §6**
Bin-for-bin LOO match to hero	North-star R&D	**No — see §6**

Path II (locked): Generative figure may bend on **LOO** `poolq_loo` **bins** (CELL 12 default) while empirical hero is the **replicated fact** on the same conditioning object — still **not** claiming pointwise match without a dedicated calibration pass.

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6. Explicitly deferred (stretch, not v1 success criteria)

Do **not** block Alex draft or side-by-side on:

1. **Bin-for-bin replication** of hero ventile draft rates from sim on `poolq_loo`.
2. **Rivanna / faithful_538** full sweeps (`outputs/simulation_sweeps/faithful_538_sweep*`) until minimal λ + K story is documented.
3. **Separate B(Q) and D(Q) estimation** on one axis.
4. **Assortativity grids**, empirical team cloning, HS binning, multiplicative rewrite.

These remain **parallel R&D** after the minimal POC bundle is frozen.

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7. Limitation sentence (Alex slide / manuscript §5)

> *The empirical stylized fact is defined on **leave-one-out pool quality** among teammates; the minimal generative proof-of-concept selects on a **score that penalizes leave-one-out viable-peer congestion** and plots binned **selection rates on the same LOO quality axis**. We do **not** claim bin-for-bin reproduction of the empirical ventile draft rates in v1; we claim that **talent-only selection is insufficient** and that **congestion in the score** can bend advancement curves in a disciplined artificial league.*

(Source: nesting note §4; shortened for slides.)

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8. Keepers vs reorder (Jul 2026 reset)

Keep: `530` pipeline, hero triplet, nesting note, 538D CELL 10–12, Pertinent Thoughts sensitivities, `alex_side_by_side_v0/` exports.

Reorder claims: Phenomenon first (hero) → minimal sim ingredient (λ) → theory prose — not "one formula does everything."

Deliverables from this reset: this memo, `lpm_hero_coefficients.txt`, generative knockout CSVs, `inverted_u_side_by_side_empirical_vs_generative.png` in `alex_side_by_side_v0/`.